

Greedy Dwarf Scenario

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Greedy Dwarf is a strategic, third-person game where players control a dwarf as they navigate through the Dragon's Lair. The goal is to gather as much treasure as possible without being caught by the dragon. The player faces the challenge of balancing speed and treasure collection, as picking up too much treasure slows the dwarf, making it harder to escape.

A key feature of the game is that the lair's layout is defined by a map file, allowing for flexible customization of the environment. This scenario focuses on the initial phase of gameplay, where the player explores the lair, collecting treasures and navigating obstacles such as walls and locked doors, without yet encountering the dragon or other characters.

Scenario "Early Exploration in the Dragon's Lair"

The player starts the game by launching it from the main menu and selecting **Start Game**. Upon starting, the system displays a brief welcome screen that explains the objective of the game: collect treasure without getting caught by the dragon. The player is informed that they must balance the amount of treasure they gather to avoid slowing down. Full instructions can be accessed from the **Instructions** section in the main menu at any time.

Once the game begins, the system loads the Dragon's Lair, as defined by a map file, and places the player in the starting area. The player controls the dwarf using the arrows key or WASD to move around. The dwarf can move in all four primary directions: North, South, East, and West.

The game accepts the following commands from the player:

- **QUIT:** Exits the game without saving progress. To exit the game, the user can either press the escape key or 'q'.
- **PAUSE:** Pausing the game brings up the pause menu, where players can resume, review instructions, or quit. To pause the game, the user should press either the space bar or 'p'.
- **INTERACT:** To pick up a treasure or use keys to open locked doors, the player must move onto a tile containing an interactive object. When this occurs, a small pop-up window will appear, displaying the object and providing instructions. To pick up the treasure or use the key, the player needs to press the 'V' key. The pop-up window will indicate the required action, such as "Press 'V' to pick up the treasure" or "Press 'V' to use the key."

If the player collects treasure, the game updates the inventory with the value and weight of the items picked up. As the dwarf accumulates weight, he will gradually slow down based on the ratio between his weight and carrying capacity. The information will be displayed in the upper right corner of the screen, indicating both the weight and the value of the treasure. The weight number will be color-coded to reflect the dwarf's load: it will appear green when the weight is low (0-39%), orange when it surpasses the medium threshold(40-79%), and red as it approaches the maximum carrying capacity(80%+). When the weight reaches the red threshold (80%), a pop-up window will be displayed, notifying the player with the message: "You are being greedy! Drop loot to move faster". If, at any point, the dwarf has accumulated weight that exceeds his carrying capacity (that is, he exceeds 100% capacity), he will instead drop the bag (and thus empty the

inventory), as it is too heavy. A direct consequence is that all the items are randomly scattered around him.

If the dwarf encounters a key, he will pick it up, and a message will be displayed: “You have picked up Key p,” where “p” corresponds to a number from 1-x (where x corresponds to the total number of doors in the lair). Each key is unique and opens only one specific door. The message will also instruct the player to search for new doors to use the key. When the dwarf finds a door and has the corresponding key, the door will open immediately. This action will consume the key, remove it from the inventory, and update the dwarf’s weight accordingly. Each key occupies a slot in the inventory with a value of 0 and a weight of 1. If the dwarf opens his inventory, he can hover over the keys to see “Key p” and identify their associated door.

As the player moves through the lair, they may encounter various obstacles. If the player tries to move through a wall or a blocked path, the system will display a message such as “You can’t go that way” or “The path is blocked”. If they encounter a locked door without the necessary key, they will see a message: “The door is locked. You need Key p” where p is the door number.

The game ends when the dwarf successfully collects a sufficient amount of treasure and exits the lair without being caught by the dragon. To achieve this, the player must locate the correct key for the exit door and use it to open the final door, leading to a successful escape. If the dragon catches the dwarf at any point, the game ends immediately, resulting in a loss. The player’s final score is based on the total value of the collected treasure and the efficiency of their escape.

Figure 1 - Diagram of the Exploration

